

The maximal energy of integral circulant graphs of order $p^{2r}3^{2s+1}$

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Abstract

Jiang and Yang determined the maximal energy of integral circulant graphs of order $n = p^{2r}q^{2s+1}$, where p, q are distinct odd primes with $q \geq 5$: the checkerboard divisor set $\{p^i q^j : i+j \text{ even}\}$ is the unique maximiser. They noted that their semidefinite method does not extend to $q = 3$. We prove the remaining case: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the checkerboard set is the unique energy maximiser for order $p^{2r}3^{2s+1}$, with the maximal energy given by the formula of Jiang and Yang. This settles the maximal-energy problem for opposite-parity exponents for all pairs of distinct odd primes. The key step is a sharp inequality for sign matrices under weighted Ramanujan transforms. We prove it with a path identity and an explicit potential function, and the same argument gives a semidefinite-free proof of the theorem of Jiang and Yang. We also state a conjecture, supported by exhaustive computation, for exponents of equal parity. The main theorem, the sign-matrix inequality and the energy formula for these graphs are formally verified in Lean 4. The proofs were found with the assistance of AI coding agents; their role is described at the end of the paper.

1 Introduction

The energy of a finite graph is the sum of the absolute values of its adjacency eigenvalues [2]. For $n \geq 2$ and a nonempty set $\mathcal{D}$ of proper divisors of n , the integral circulant graph $\text{ICG}_n(\mathcal{D})$ has vertex set $\mathbb{Z}/n\mathbb{Z}$, two distinct vertices u, v being adjacent when $\gcd(u - v, n) \in \mathcal{D}$. These are the circulant graphs with integral spectrum [8]. The problem of maximising the energy of $\text{ICG}_n(\mathcal{D})$ over $\mathcal{D}$ for fixed n has been solved for prime powers [6] and studied for other orders; see the introduction of [3] for a survey. Roldán [5] conjectured that for $n = p^2 q^3$ the checkerboard set $\{1, p^2, pq, q^2, p^2 q^2, pq^3\}$ is the unique maximiser. Jiang and Yang [3] proved much more.

Theorem 1 (Jiang–Yang [3, Theorem A]). *Let p, q be distinct odd primes with $q \geq 5$, let $r \geq 1$, $s \geq 0$ and $n = p^{2r}q^{2s+1}$. Among all nonempty sets of proper divisors of n , the energy of $\text{ICG}_n(\mathcal{D})$ is uniquely maximised by*

$$\mathcal{D}_{r,s}^* = \{p^i q^j : 0 \leq i \leq 2r, 0 \leq j \leq 2s+1, i+j \text{ even}\},$$

and the maximal energy is $\frac{1}{2}[n + d_{2r}(p) d_{2s+1}(q)]$, where $d_k(x) = (2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j} (j+1)x^j$.

They remark that the case $q = 3$ is not covered, because their odd-exponent semidefinite argument fails at $x = 3$ [3, Remark 4.2]. Our main result removes this restriction.

Theorem 2. *The conclusion of Theorem 1 holds for $q = 3$: for every prime $p \geq 5$ and all $r \geq 1$, $s \geq 0$, the set $\mathcal{D}_{r,s}^*$ is the unique energy maximiser among nonempty sets of proper divisors of $n = p^{2r}3^{2s+1}$, and the maximal energy is $\frac{1}{2}[n + d_{2r}(p) d_{2s+1}(3)]$.*

Together with Theorem 1 (which allows $p = 3$ on the even side), Theorem 2 determines the maximal energy and the unique maximiser for every order $p^{\text{even}}q^{\text{odd}}$ with p, q distinct odd primes. The case $r = s = 1$ was treated earlier. Park [4] proved Roldán's conjecture for all distinct odd primes. Schreib [7] proved the case $q = 3$ and combined it with [3] to obtain the conjecture. An arithmetic certificate for $n = 27p^2$ was announced in [9]. These works concern only the exponents $(2, 3)$, and we are not aware of any earlier treatment of the case $q = 3$ for general r and s .

Our proof replaces the semidefinite part of [3] by Theorem 9 below, an inequality for sign matrices that holds for all exponents and parities and all real $p \geq 5$, $q \geq 3$. By symmetry it also yields Theorem 1 without semidefinite certificates (Remark 10). In Section 8 we record a conjecture, supported by exhaustive computation, for exponents of equal parity.

2 Weighted Ramanujan transforms

We use the notation of [3]. For real $x > 1$, an integer $k \geq 1$ and $0 \leq i, j \leq k$, let $T_k(x) = (t_{ij})$ be the symmetric matrix with

$$t_{ij} = \begin{cases} 0, & i + j \leq k - 2, \\ -x^{k-1}(x-1), & i + j = k - 1, \\ x^{2k-i-j-2}(x-1)^2, & i + j \geq k, \end{cases} \quad (i, j < k),$$

$t_{ik} = t_{ki} = x^{k-i-1}(x-1)$ for $i < k$, and $t_{kk} = 1$. When x is prime, $t_{ij} = \varphi(x^{k-i}) c_{x^{k-j}}(x^i)$, where $c_m(u)$ denotes Ramanujan's sum [3, (2.2)–(2.4)]. The row sums are $T_k(x)\mathbf{1} = x^k e_k$ [3, (2.5)], where e_k is the last standard basis vector. For a real matrix Z let $\|Z\|_1$ be the sum of the absolute values of its entries.

Proposition 3 ([3, Proposition 2.2]). *Let $p \neq q$ be primes, $a, b \geq 1$, $n = p^a q^b$, and let $X \in \{0, 1\}^{(a+1) \times (b+1)}$ have $x_{ij} = 1$ exactly when $p^i q^j \in \mathcal{D}$. Then $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|T_a(p)XT_b(q)^\top\|_1$.*

Let $s_k = (1, -1, 1, \dots, (-1)^k)^\top$ and define $d_k(x) = \|T_k(x)s_k\|_1$, and put $d_0(x) = 1$. By Remark 6 below, $d_k(x)$ agrees with the closed form in Theorem 1 for all real $x > 2$; see also [3, (3.6)].

3 A path identity

For $w \in \mathbb{R}^{k+1}$ define $z_{k-1} = w_k$ and, for $j = k-1, k-2, \dots, 0$,

$$z_{j-1} = \frac{z_j + (x-1)w_j}{x}. \quad (1)$$

Thus z_j is a convex combination of $w_{j+1}, \dots, w_k$, and each z_j is a linear function of w .

Lemma 4. *For $w \in \mathbb{R}^{k+1}$ we have*

$$\|T_k(x)w\|_1 = x^k \left(|z_{-1}| + \sum_{j=0}^{k-1} |z_{j-1} - z_j| \right).$$

Proof. Unwinding (1) gives $z_j = x^{-(k-j-1)} \sum_{i>j} \varphi(x^{k-i}) w_i$ with $\varphi(x^0) = 1$ and $\varphi(x^e) = x^{e-1}(x-1)$. The last row of $T_k(x)$ is $(\varphi(x^{k-i}))_i$, so $(T_k(x)w)_k = x^k z_{-1}$. For $u < k$ put $j = k-1-u$. Row u has entry $-x^{k-1}(x-1)$ in column j , entries $x^j(x-1)\varphi(x^{k-i})$ in columns $i > j$, and zeros elsewhere. Hence $(T_k(x)w)_u = x^{k-1}(x-1)(z_j - w_j)$, while $z_{j-1} - z_j = \frac{x-1}{x}(w_j - z_j)$. $\square$

4 The one-dimensional potential

For $x > 2$ let $\mu_{-1} = 1$, $\mu_j = (x - 1 - \mu_{j-1})/x$ for $j \geq 0$, and $c_j = (x - 1)(1 + \mu_{j-1})/x$. By induction, $\mu_j \in [\frac{x-2}{x}, 1]$ for all j .

Lemma 5. *Let $x > 2$, let $y \in \{\pm 1\}^{k+1}$, and let z_j be defined by (1) with $w = y$. Then $\text{sgn } z_j = y_{j+1}$ and $\frac{x-2}{x} \leq |z_j| \leq 1$ for $-1 \leq j \leq k-1$, and*

$$\frac{\|T_k(x)y\|_1}{x^k} = \sum_{j=0}^{k-1} c_j + \mu_{k-1} - \sum_{j \in \text{NA}(y)} 2\mu_j |z_j|, \quad \text{NA}(y) = \{0 \leq j \leq k-1 : y_j = y_{j+1}\}.$$

Consequently $\|T_k(x)y\|_1 \leq d_k(x)$, with equality exactly when $y = \pm s_k$, and $d_k(x) = x^k(\sum_j c_j + \mu_{k-1})$. Moreover $(T_k(x)s_k)_k > 0$.

Proof. We have $z_{k-1} = y_k$, and since $x - 1 > 1 \geq |z_j|$, the sign of $z_{j-1} = (z_j + (x-1)y_j)/x$ is y_j and $|z_{j-1}| = (x-1 \pm |z_j|)/x \in [\frac{x-2}{x}, 1]$. Put $f_j = |z_{j-1} - z_j| + \mu_{j-1}|z_{j-1}| - \mu_j|z_j|$. If $y_j = -\text{sgn } z_j$, then $|z_{j-1} - z_j| = \frac{x-1}{x}(1 + |z_j|)$ and $|z_{j-1}| = \frac{x-1-|z_j|}{x}$; using $x\mu_j = x-1-\mu_{j-1}$ one finds $f_j = c_j$. If $y_j = \text{sgn } z_j$, the same computation with $|z_{j-1} - z_j| = \frac{x-1}{x}(1 - |z_j|)$ and $|z_{j-1}| = \frac{x-1+|z_j|}{x}$ gives $f_j = c_j - 2\mu_j|z_j|$. Summing, the terms $\mu_j|z_j|$ telescope, and Lemma 4 gives the identity, since $\mu_{-1} = 1$ and $|z_{k-1}| = 1$. Every $j \in \text{NA}(y)$ contributes a positive term $2\mu_j|z_j|$, and $\text{NA}(y) = \emptyset$ exactly for $y = \pm s_k$. Finally $(T_k(x)s_k)_k = x^k z_{-1}$ and $\text{sgn } z_{-1} = (s_k)_0 = 1$. $\square$

Remark 6. Put $\delta_k(x) = x^k \mu_{k-1}$. Solving the recursion for μ gives $\delta_k(x) = (x^k(x-1) + 2(-1)^k)/(x+1)$. Since the potential does not depend on k , Lemma 5 gives $d_{k+1}/x^{k+1} - d_k/x^k = c_k + \mu_k - \mu_{k-1} = 2\mu_k$, that is, $d_{k+1}(x) = x d_k(x) + 2\delta_{k+1}(x)$, with $d_0 = 1$. The closed form $(2k+1)x^k + 4 \sum_{j=0}^{k-1} (-1)^{k-j}(j+1)x^j$ satisfies the same recursion and initial value, so it equals $d_k(x)$ for all $k \geq 0$.

5 A two-variable inequality

Lemma 7. *Let $P \geq \mu$ and $0 \leq \mu \leq 1$, and put $h(A, B) = P|A - B| + \mu|B + PA| - (P - \mu)|B| - P(1 + \mu)|A|$ for real A, B . Then $h(A, B) \leq 2\mu \max\{0, |B| - P|A|\}$.*

Proof. As $h(-A, -B) = h(A, B)$ we may assume $A \geq 0$. If $0 \leq B \leq A$ then $h = -2(P - \mu)B \leq 0$. If $B > A$ then $h = 2(\mu B - PA) \leq 2\mu(B - PA)$ because $\mu \leq 1$. If $-PA \leq B < 0$ then $h = 0$. If $B < -PA$ then $h = 2\mu(|B| - PA)$. $\square$

Lemma 8. *Let $q \geq 3$, $b \geq 1$, $N = T_b(q)$, $F(w) = \|Nw\|_1$ and $D = d_b(q)$. Let $P \geq 4$ and $\frac{3}{5} \leq \mu \leq 1$. For all $y \in \{\pm 1\}^{b+1}$ and $\zeta \in [-1, 1]^{b+1}$,*

$$\Lambda(y, \zeta) := P F(y - \zeta) + \mu F(\zeta + Py) - (P - \mu) F(\zeta) \leq P(1 + \mu) D,$$

with strict inequality unless $y = \pm s_b$.

Proof. Let $\xi_j(w)$ be the path (1) with $x = q$, $k = b$, and put $\tau_{-1}(w) = \xi_{-1}(w)$, $\tau_j(w) = \xi_{j-1}(w) - \xi_j(w)$ for $0 \leq j < b$. These are linear, and $F(w) = q^b \sum_j |\tau_j(w)|$ by Lemma 4. With $A_j = \tau_j(y)$ and $B_j = \tau_j(\zeta)$,

$$\Lambda/q^b = P(1 + \mu)F(y)/q^b + \sum_j h(A_j, B_j).$$

By Lemma 5 (with $x = q$ and potential μ_j^q), it suffices to show $\sum_j h(A_j, B_j) \leq \sum_{j \in \text{NA}(y)} 2P(1 + \mu)\mu_j^q x_j$, where $x_j = |\xi_j(y)| \in [\frac{q-2}{q}, 1]$, with strict inequality when $\text{NA}(y) \neq \emptyset$. Since $\xi_j(\zeta)$ is a convex combination of entries of ζ , we have $|B_{-1}| \leq 1$ and $|B_j| = \frac{q-1}{q}|\zeta_j - \xi_j(\zeta)| \leq \frac{2(q-1)}{q}$. Also $|A_{-1}| \geq \frac{q-2}{q}$, and $|A_j| = \frac{q-1}{q}(1 + x_j)$ for $j \notin \text{NA}(y)$, $|A_j| = \frac{q-1}{q}(1 - x_j)$ for $j \in \text{NA}(y)$. We use Lemma 7, noting $P \geq 4 > \mu$.

- $j = -1$: $P|A_{-1}| \geq \frac{4(q-2)}{q} \geq \frac{4}{3} > |B_{-1}|$, so $h \leq 0$.
- $j \geq 0, j \notin \text{NA}(y)$: $P|A_j| \geq \frac{8(q-1)^2}{q^2} \geq \frac{2(q-1)}{q} \geq |B_j|$, so $h \leq 0$.
- $j \in \text{NA}(y)$: it suffices that $\mu|B_j| < P\beta(x_j)$, where $\beta(t) = \mu \frac{q-1}{q}(1-t) + (1+\mu)\mu_j^q t$. As β is affine it suffices to check the endpoints. For $q = 3$, $\beta(1) \geq \frac{8}{5} \cdot \frac{1}{3}$ and $\beta(\frac{1}{3}) \geq \frac{3}{5} \cdot \frac{4}{9} + \frac{8}{5} \cdot \frac{1}{9} = \frac{4}{9}$, so $P\beta \geq \frac{16}{9} > \frac{4}{3} \geq \mu|B_j|$. For general $q \geq 3$ the same comparison reduces to $P\beta(1) \geq \frac{32}{5} \cdot \frac{q-2}{q} > \frac{2(q-1)}{q}$ and $\frac{22}{5}q^2 - \frac{94}{5}q + \frac{104}{5} > 0$.

Summing over j proves the claim. $\square$

6 The sign-matrix inequality

Theorem 9. Let $p \geq 5$ and $q \geq 3$ be real, $a, b \geq 1$, $M = T_a(p)$ and $N = T_b(q)$. For every $Y \in \{\pm 1\}^{(a+1) \times (b+1)}$,

$$\|MYN^\top\|_1 \leq d_a(p) d_b(q),$$

with equality exactly when $Y = \pm s_a s_b^\top$.

Proof. Let $\bar{y}_0, \dots, \bar{y}_a$ be the rows of Y . Define $\zeta_{a-1} = \bar{y}_a$ and $\zeta_{k-1} = (\zeta_k + (p-1)\bar{y}_k)/p \in [-1, 1]^{b+1}$. Column v of $MYN^\top$ is $Mw^{(v)}$ with $w_i^{(v)} = (N\bar{y}_i)_v$, and by linearity the path of $w^{(v)}$ is $((N\zeta_k)_v)_k$. Applying Lemma 4 with $x = p$ to every column and summing,

$$\|MYN^\top\|_1 = p^a \left(F(\zeta_{-1}) + \sum_{k=0}^{a-1} F(\zeta_{k-1} - \zeta_k) \right), \quad F(w) = \|Nw\|_1.$$

Let μ_k be the potential of Lemma 5 for $x = p$ and put $f_k = F(\zeta_{k-1} - \zeta_k) + \mu_{k-1}F(\zeta_{k-1}) - \mu_k F(\zeta_k)$. Telescoping gives $\|MYN^\top\|_1/p^a = \sum_k f_k + \mu_{a-1}F(\bar{y}_a)$. Since $\zeta_{k-1} - \zeta_k = \frac{p-1}{p}(\bar{y}_k - \zeta_k)$ and $p\mu_k = p-1 - \mu_{k-1}$, we get $pf_k = \Lambda(\bar{y}_k, \zeta_k)$ with $P = p-1 \geq 4$ and $\mu = \mu_{k-1} \in [\frac{p-2}{p}, 1] \subseteq [\frac{3}{5}, 1]$. Lemma 8 gives $f_k \leq c_k d_b(q)$, and Lemma 5 gives $F(\bar{y}_a) \leq d_b(q)$; both are strict unless the row involved is $\pm s_b$, and $\mu_{a-1} > 0$. Hence $\|MYN^\top\|_1 \leq p^a(\sum_k c_k + \mu_{a-1})d_b(q) = d_a(p)d_b(q)$. If equality holds, every row of Y is $\pm s_b$, so $Y = g s_b^\top$ with $g \in \{\pm 1\}^{a+1}$ and $\|MYN^\top\|_1 = \|Mg\|_1 d_b(q)$; by Lemma 5 this equals $d_a(p)d_b(q)$ only for $g = \pm s_a$. $\square$

Remark 10. Since $\|MYN^\top\|_1 = \|NY^\top M^\top\|_1$, the roles of the two factors may be exchanged, so Theorem 9 holds whenever $\min(p, q) \geq 3$ and $\max(p, q) \geq 5$. In particular it contains the sign-matrix inequality [3, Lemma 4.1] used for Theorem 1, and the deduction below then proves Theorem 1 as well. The semidefinite argument of [3] does not extend to $x = 3$ [3, Remark 4.2]; the proof above instead uses the integrality of the signs on both factors.

7 Proof of Theorem 2

Let $a = 2r$, $b = 2s + 1$, $M = T_a(p)$, $N = T_b(3)$, let X be the divisor matrix of $\mathcal{D}$ (so $x_{ab} = 0$), and let $Y = 2X - J$ with J the all-ones matrix. By Proposition 3 and $M\mathbf{1} = p^a e_a$, $N\mathbf{1} = 3^b e_b$,

$$2\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \|MJN^\top + MYN^\top\|_1 = \|n e_a e_b^\top + MYN^\top\|_1 \leq n + d_a(p) d_b(3),$$

by Theorem 9. For the checkerboard set, $Y^* = s_a s_b^\top$, $x_{00}^* = 1$, and $x_{ab}^* = 0$ because $a + b$ is odd, so $\mathcal{D}_{r,s}^*$ is an admissible nonempty set. The (a, b) entry of $MY^*N^\top = (M s_a)(N s_b)^\top$ is positive by Lemma 5, so it does not cancel against n , and $2\mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*)) = n + \|M s_a\|_1 \|N s_b\|_1 = n + d_a(p) d_b(3)$. If $\mathcal{E}(\text{ICG}_n(\mathcal{D})) = \mathcal{E}(\text{ICG}_n(\mathcal{D}_{r,s}^*))$, then $\|MYN^\top\|_1 = d_a(p) d_b(3)$, so $Y = \pm s_a s_b^\top$ by Theorem 9; since $y_{ab} = -1 = (s_a s_b^\top)_{ab}$, we get $Y = Y^*$ and $\mathcal{D} = \mathcal{D}_{r,s}^*$. $\square$

8 Exponents of equal parity

If $a + b$ is even, the sign matrix $s_a s_b^\top$ has entry $+1$ at (a, b) and violates the proper-divisor condition. The matrix $-s_a s_b^\top$, which corresponds to the set $\{p^i q^j : i + j \text{ odd}\}$, is admissible, and it attains the bound of Theorem 9. However, the (a, b) entry of $MYN^\top$ is then negative and cancels against n in the argument of Section 7. So the energy bound $\frac{1}{2}[n + d_a(p) d_b(q)]$ is not attained, and the maximiser changes. Let δ_k be as in Remark 6.

Conjecture 11. *Let p, q be distinct odd primes, $a, b \geq 1$ with $a + b$ even, and $n = p^a q^b$. The maximal energy of an integral circulant graph of order n is*

$$\frac{1}{2}[n + d_a(p) d_b(q)] - \delta_a(p) \delta_b(q),$$

attained exactly by the sets $\{p^i q^j : i + j \text{ odd}\}$ and $\{p^i q^j : i + j \text{ even}\} \setminus \{n\}$.

For $a = b = 1$ the formula gives $4(p - 1)(q - 1)$. An exhaustive search over all divisor sets confirms the conjecture for ten ordered pairs of odd primes and each of the exponent pairs $(1, 1)$, $(2, 2)$, $(1, 3)$, $(3, 1)$, $(3, 3)$, $(2, 4)$, $(4, 2)$, $(4, 4)$, $(1, 5)$ and $(3, 5)$. We have not checked whether small cases are covered by the literature.

9 Verification

The proofs above are complete; the following computations are independent cross-checks.

- An exhaustive search over all divisor sets, in exact integer arithmetic, confirms Theorem 2, including the value of the maximum, in the following cases:
 - exponent pairs $(4, 1)$, $(6, 1)$, $(8, 1)$, $(2, 5)$, $(4, 3)$, for the primes $5 \leq p \leq 31$;
 - $(10, 1)$, $(12, 1)$, $(2, 7)$, $(6, 3)$, for $p \in \{5, 7, 11, 13, 17\}$;
 - $(14, 1)$, $(4, 5)$, $(2, 9)$, for $p \in \{5, 7, 11\}$ (the largest case has 2^{29} divisor sets).
- Theorem 9 was checked in exact arithmetic over all sign matrices of every shape with $(a + 1)(b + 1) \leq 25$, for $(p, q) = (5, 3)$, $(7, 3)$ and $(5, 4)$.
- Each lemma was tested separately in exact arithmetic: by adversarial searches for small b , and by structured random tests of Lemma 8 up to $b = 20$.
- Proposition 3 was compared, in floating-point arithmetic, with spectra of the actual circulant matrices. The fast Fourier transform was used for all divisor sets of the orders $n = 75, 147, 363, 675, 1323, 1875$ and 6075 . Dense diagonalisation was used for all divisor sets when $n \leq 675$ and for samples of them when $n = 1323, 1875$ and 6075 .

Formal verification. Theorems 2 and 9 are verified in Lean 4 (version 4.33.1, Mathlib commit 0df444a3). The development starts from the definition of the graph and of its energy:

- the adjacency matrix of $\text{ICG}_n(\mathcal{D})$ has entry 1 exactly when $\gcd((j - i) \bmod n, n) \in \mathcal{D}$;
- the energy is the sum of the absolute values of its eigenvalues, as given by Mathlib's spectral theorem.

It proves:

- Lemmas 4–8 and Theorem 9 for real $p \geq 5$, $q \geq 3$ and all $a, b \geq 1$, together with the exchanged form of Remark 10;

- Proposition 3 for all distinct primes p, q and all exponents, from the eigenvalues of circulant matrices;
- the conclusion of Theorem 2 and the value of the maximal energy, with $d_k(x)$ defined as $\|T_k(x)s_k\|_1$;
- the conclusion of Theorem 1 for all pairs of distinct odd primes, which includes Theorem 2 and the case $q \geq 5$.

The whole development was rebuilt from a clean directory. For every theorem, the command `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`; no native evaluation is used. The code and logs are available at [1].

Tool and computational resource disclosure

This work was carried out between 25 and 26 September 2026 in a workflow directed by the author, using the AI coding agents Anthropic Claude Code (model Claude Opus 5.5) and OpenAI Codex (desktop application; models `gpt-6-astra` and `gpt-6-sol`). The author chose the research programme and directed the agents.

- A Claude Code agent found the proof of Theorem 9 and of Theorem 2, formulated Conjecture 11, and wrote the verification programs.
- Another Claude Code agent wrote the Lean formalisation.
- The proof was checked line by line by a second Claude Code agent and by Codex, and the agents searched the literature.

Claude Code drafted this text. The programs and logs are available at [1]. The author is not a professional mathematician and has read the paper. The correctness of the results rests on the Lean formalisation described in Section 9 and on the independent checks listed there. In addition, the arguments were checked by AI agents of two different model families, and the formal statements were reviewed by an agent that did not write them. AI systems are not authors of this paper; the author takes full responsibility for its content.

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